

Course (TY02011): (SYSTEMIC VETERINARY PATHOLOGY)

1. General information

- Term: 05
- Credits: **Total credits 03 (Lecture: 2.5 – Practice: 0.5)**
- **Self-study: 09** credits
- Credit hours for teaching and learning activities: 45 hrs
- Self-study: 135 hrs.
- Department conducting the course:
 - Department: Pharmacology, Internal Medicine, and Diagnostic
 - Faculty: Veterinary Medicine
- Kind of the course:

Foundation ☐		Fundamental ☒		Option 1 ☐	
Compulsory	Elective	Compulsory	Elective	Compulsory	Elective
☐	☐	☒	☐	☐	☐

- Prerequisite course(s): TY02020_ General Veterinary Pathology

2. Course objectives and expected learning outcomes

* *Course objectives:*

- Knowledge: Course provides for students with basic knowledge of pathological lesions from different causes and in body systems to help veterinarians have ability to diagnosis and distinguish animal diseases quickly and accurately.
- Skills: Course provides students with skills to distinguish major microscopic lesions
- Attitude: Course provides students to have competence in the trained pathology expertise; have the ability to self-direct, self-study, accumulate knowledge and experience to improve professional qualifications in pathology; be able to make conclusions on common medical professional issues and some complex issues in veterinary pathology; have the ability to evaluate and improve professional activities at a moderate level.

* *Course expected learning outcomes*

Notation	Course expected learning outcomes (CLOs) After successfully completing this course, students are able to	Program performance criteria (PPC)
Knowledge		
CLO1	Applying knowledge of pathological characteristics on organs and systems to prevent and treat animal diseases	3.3
Skills		
CLO2	Proficiently perform clinical and non-clinical skills, diagnostic procedures and pathological treatment in organs and system of animal	8.1

CLO3	Using technology and machinery and equipment for diagnosis, prevention, treatment and management of animal diseases to achieve the goals	10.1
CLO4	Applying research results of pathology on organs, systems and animals to find problems	10.2, 11.1
Attitude		
CLO5	Comply with the law and professional ethics in prevention and treatment of animal disease	12.2

3. Course description

TY02011. Systemic veterinary pathology. (3 credits: 2.5-0.5-6). Content: Pathology of the circulatory system; Pathology of the respiratory system; Pathology of the digestive system; Pathology of the urinary system; Pathology of the genito-urinary system; Pathology of chronic infectious diseases caused by bacteria; Pathology of acute infectious diseases caused by bacteria; Pathology of viral diseases. **Teaching methods:** Students listen to lectures in class and practice in experimental room, combined with self-study, suggested documents and exchange with friends and lecturers. **Assessment method:** Attendance: 10%, midterm: 30%, final exam: 60%. Pre-course: Veterinary Pathology 1

4. Teaching and learning & assessment methods

CLOs	CLO1	CLO2	CLO3	CLO4	CLO5
Teaching methods					
Lecturing	x				
Teaching through practical work			x	x	x
Assessment					
Rubric 1. Attendance (10 %)					x
Rubric 2. Practical (30 %)			x	x	
Rubric 4. Final exam (60 %)		x	x	x	

5.

Student tasks

- Attendance: All students will be expected to attend at least 70% of lecturing.
- Preparation for the lecture: All students take notes on these content outlines before each class
- Practices: All students must fully participate of the subject practice sessions and ensure having personal protective equipment for subject practice
- Final exam: All students must fully participate in the subject practice sessions, group presentations, midterm exams and final exams.

6. Textbooks and references

* *Textbooks/Lecture Notes:*

1. Nguyen Huu Nam, Nguyen Thi Lan, Bui Tran Anh Dao (2016). Veterinary Pathology 2. University of Agricultural Publisher
2. Nguyen Huu Nam, Nguyen Thi Lan Bui Tran Anh Dao (2015). Curriculum practice veterinary pathology. University of Agricultural Publisher

*** Additional references:**

1. Jaap E. Van Dijk, Erik Gruys, Johan M.V.M. Mouwen (2007). Color Atlas of Veterinary Pathology (Second Edition)
2. José Antonio García de Jalón Ciercoles, Marcelo de las Heras Guillamón (2012). A guide to necropsy diagnosis in swine pathology.

7. Course outline

Week	Content	Course expected learning outcomes
1	Chapter 1. Vascular system pathology	
	A/ Main contents: (03 hours) Theory: (03 hours) Outline of vascular system Common forms of pathological lesions in the heart	K1, K2
	B/ Self - learning contents: (09 hours) Lesions in the vascular system	K1, K2, K4
2	Chapter 2. Respiratory system Pathology	
	A/ Main contents: (05 hours) Theory: (05 hours) Summarizes the basic structure of respiratory Lesions in bronchial Lesions in lung	K1, K2
	B/ Self - learning contents: (15 hours) Respiratory Dysfunction Pulmonary edema	K1, K2, K4
3	Chapter 3. Digestive system pathology	
	A/ Main contents: (8 hours) Theory: (05 hours) -Dysfunctions of the digestive system -Lesions of the digestive system Practical (03 hours) - Outpatient Endocarditis -Pericardial Cardiology -Pulmonary Apse -Hemorrhagic in lungs - Tuberculosis	K1, K2, K3

	B/ Self - learning contents: (24 hours) Outline of digestion system	K1, K2, K4
4	Chapter 4. Urological system pathology	
	A/ Main contents: (03 hours) Theory: (03 hours) Common forms of kidney inflammation Lesions in the bladder and urethra	K1, K2
	B/ Self - learning contents: (09hours) - Kidney outline -Congenital abnormalities and diseases of the kidney cyst	K1, K2, K4
5	Chapter 5. Reproductive system pathology	
	A/ Main contents: (6hours) Theory: (3 hours) Dysfunction of the genitourinary system Morphological lesions of the genitourinary system Forms of uterine inflammation Forms of testicles inflammation Forms of mastitis inflammation Practical (03 hours)	K1, K2, K3
	B/ Self - learning contents: (18 hours) - Acute glomerulonephritis - inflammation of chronic interstitial nephritis	K1, K2, K4
6	Chapter 6. Pathology of chronic infectious disease caused by bacteria	
	A/ Main contents: (5 hours) Theory: (3 hours) - Mycoplasmosis - Leptospirosis - Tuberculosis - Fungus disease - Brucellosis Practical (2 hours) - Hemorrhagic lymph nodes - Bronchopneumonia	K1, K2, K3
	B/ Self - learning contents: (15 hours) Biological characteristic of bacteria	K1, K2, K4
7	Chapter 7. Pathology of acute infectious disease caused by bacteria	
	A/ Main contents: (4 hours) Theory: (4 hours) - Anthrax disease - Pasteurellosis (Poultry, Pigs, Buffalo - Beef) - Salmonellosis (Poultry, Pigs, Buffalo - Beef)	K1, K2, K3

	- Erysipelas - Escherichia coli	
	B/ Self - learning contents: (12 hours) Biological characteristic of bacteria	K1, K2, K4,
	Chapter 8. Pathology of infectious disease caused by viruses	
8	A/ Main contents: (6 hours) Theory: (6 hours) - Foot and Mouth Disease - Classical swine fever - African swine fever - TGE - Aujeszky - Patients errors puppies - Hepatitis viral - Influenza in the species - Newcastle disease - Gumboro disease - Cholera duck - Marek disease - Leukosis - Infectious Diseases bronchitis - Circovirus in pig	K1, K2, K3
	B/ Self - learning contents: (18 hours) Biological characteristic of virus	K1, K2, K4
	Chapter 9. Pathology of toxicological diseases	
9	A/ Main contents: (5 hours) Theory: (5 hours) Pathology of toxicological diseases	K1, K2, K3
	B/ Self - learning contents: (15 hours) Poisoning due to the use of veterinary drugs	K1, K2, K4